COM SCI 263: Natural Language Processing
Final Project

Overview

This course project is designed to immerse you in the authentic experience of conducting research within the field of Natural Language Processing (NLP), enhance your understanding of NLP tasks and principles, sharpen your problem-solving skills, and stimulate innovation. Offering a curated selection of four project topics, **the course also encourages creativity and individual initiative by allowing students the flexibility to propose their own topics**. Note that if you decide to propose your own research topic, please write a proposal with details similar to the provided project descriptions. Through this hands-on project, you will navigate the research process from designing steps like hypothesis formulation and experimental setup, to implementation steps like experiment execution, culminating in a comprehen-sive report and a presentation of your findings. In summary, this project provides an opportunity to apply theoretical knowledge to practical challenges, preparing you for future endeavors in NLP and related fields.

Submission

There are several important milestones for the final project. Please submit each of them to BruinLearn following the description below:

1. **Project planning report due: 4/17/2026 at 11:59PM. (10%)**
   - Teaming information
   - The research topic you choose to work on (can be your own topic)
   - The distribution of work among teammates
2. **Project midterm report due: 5/4/2026 at 11:59PM. (20%)**
   - Introduction to the problem
   - Literature Review and baseline model selection
   - Description of your approach
   - Project progress until this point
   - Weekly plan and distribution of work for the subsequent weeks.
3. **Course project final presentation starts: 6/01/2026 (30%)**
4. **Project final report and code due: 6/08/2026 at 11:59PM. (40%)**
   - Upload your code and report to BruinLearn.
     - Please include a README including
       - How to set up your code environment?
       - How to obtain the data you use?
       - How to reproduce your results? e.g. the training scripts (if applicable) and the inferences scripts.
     - For closed-sourced LLMs such as ChatGPT, we expect you to write code to call their API, as well as playing with different inference setups (e.g., different algorithms and/or hyperparameters) instead of using the web GUI.

  - You are anticipated to write code for result analysis, generating figures and plots, etc.
- A 4-8 pages research paper-like writeup, preferably using this template.
- Extend your midterm report to include more details about your approach.
- Dataset and evaluation framework.
- Experimental results with quantitative and qualitative analysis of your method vs. baselines.

Project Idea 1: Evaluating the Limitations of Generative Models in Complex Applications

The rapid advancement of generative models, particularly Large Language Models (LLMs) and Vision-Language Models (VLMs), has opened up a wide array of opportunities for downstream applications. However, there remain critical challenges when deploying these models in complex downstream applications, such as scientific reasoning, code generation, image/video captioning, and spatial/3D reasoning. In this project, you will select one such challenging application (be creative!) and design a series of evaluation metrics and protocols that precisely reflect the limitations of current generative models in that domain. Your goal is to create an evaluation framework that not only highlights these shortcomings but also sheds light on potential root causes and areas for improvement.

Tasks:

1. Choose a downstream application of LLMs/VLMs that you want to study.
   (a) Some example options: Scientific reasoning, Code generation, Video captioning, Image captioning, Spatial reasoning, 3D reasoning.
   (b) You are encouraged to propose other applications in your project proposal if they align with the focus of generative models.
2. Select 1-2 LLMs/VLMs for evaluation.
   (a) Options include ChatGPT or other open-source models like LLaMA, LLaVA, etc.
3. Design or identify a test dataset
   (a) Create a dataset with **at least 50** data entries (inputs + ground truth answers) to evaluate model responses.
   (b) Example: For scientific reasoning, create a dataset of complex questions requiring multi-step problem-solving, where models are more likely to struggle with accuracy or logical consistency. For example, you could ask models to generate explanations for why a specific scientific phenomenon occurs, focusing on questions that require both factual knowledge and deeper reasoning.
4. Design methods and automated metrics to evaluate model's performances.
   (a) For video captioning, metrics might include accuracy in describing the content, coherence of the narrative, and the general fluency of the captions.
   (b) For each metric you design, explain why it is essential for evaluating models in the chosen application. For instance, in video captioning, temporal accuracy is critical because the model needs to describe events in sequence, while coherence ensures the generated text forms a logical narrative.
5. Apply the designed metrics to assess model's performance. Report the outcomes clearly, focusing on areas where the model excels or struggles.
6. (Optional) Perform human annotation on model outputs and compare human judgment with metric outcomes.
   - Human annotation can be used to justify the design of evaluation methods.
7. (Optional) Propose a preliminary solution to improve model's performance on the designed testing dataset.
   (a) Provide a simple, exploratory suggestion that sheds light on how the model's performance might be improved. This doesn't need to be complex, but should point to a possible research direction.
8. Prepare the project final presentation.
9. Draft a report with both quantitative and qualitative results.

Suggested Readings:

1. A Survey on Evaluation of Large Language Model [4]
2. Program Synthesis with Large Language Models [2]
3. What's "up" with vision-language models? Investigating their struggle with spatial reasoning [11]
4. MathVista: Evaluating Mathematical Reasoning of Foundation Models in Visual Contexts [18]
5. EMScore: Evaluating Video Captioning via Coarse-Grained and Fine-Grained Embedding Matching [24]
6. ScienceWorld: Is your Agent Smarter than a 5th Grader? [30]
7. Davidsonian Scene Graph: Improving Reliability in Fine-grained Evaluation for Text-to-Image Generation [5]
8. Winoground: Probing Vision and Language Models for Visio-Linguistic Compositionality [26]

Project Idea 2: Cross-Cultural Social Acceptability Classification: Enhancing Model Performance Across Diverse Contexts via Human Feedback

Social acceptability classification plays a vital role in identifying instances of harm or discomfort for individuals within a given context. However, existing approaches to social acceptability classification often reflect narrow cultural perspectives, overlooking the cultural variability in social norms and acceptable behaviors. This project aims to create a flexible social acceptability classification (SAC) model that adapts based on human feedback, adjusting responses according to diverse cultural contexts. The goal is to investigate whether models can adjust their predictions when provided with human feedback or specific cultural policies and, further, how to steer models to utilize this feedback effectively. As a reference, a dataset to test **general model accuracy without culture context** can be found at dataset1 , dataset2.

Tasks:

1. Identify culturally dependent data points: Develop a comprehensive dataset with **at least 50** data entries. Each sample in the dataset should comprise of:
   (a) **An interaction**: a chat with an LLM.
   (b) **Ground truth labels**: labels indicating the acceptability within a particular cultural context.
   (c) **Cultural context**: manually create the culture-specific norms that influence the acceptability of the interaction. It can be some general principles for a given culture, or detailed explanations for individual data points.
2. Baseline Model Evaluation: Establish a baseline performance of 1-2 existing social acceptability classification models using the dataset.
   (a) Evaluate the model on the dataset **without** any human feedback or additional cultural context. Measure the model's performance in predicting social acceptability across different cultural contexts based on pre-trained knowledge.
   (b) Analyze the results: Identify areas where the model struggles to account for cultural variability and mismatches between its responses and the ground truth labels.
3. Baseline Model with Human Feedback:
   (a) Introduce human-defined policies or feedback into the evaluation. Provide the model with explicit instructions or cultural guidelines in the prompt that explain what is considered acceptable in a particular culture.
   (b) Evaluate how the model adjusts its predictions based on this feedback. Measure how well it adapts to different cultural norms and corrects its responses.
4. Develop novel methodologies for enhancing model adaptability to diverse cultural contexts
   (a) Explore techniques to enhance the model's ability to adapt to diverse cultural contexts, techniques include but not limited to: in-context learning, prompt engineering, retrieval-augmented generation (RAG), reinforcement learning with human/AI's feedback (RLHF/RLAIF).
5. Prepare the project final presentation.
6. Draft a report with quantitative and qualitative analysis.

Suggested Readings:

1. NLPositionality: Characterizing Design Biases of Datasets and Models [22]
2. The Socio-Moral Image Database (SMID): A novel stimulus set for the study of social, moral and affective processes [7]
3. SALAD-Bench: A Hierarchical and Comprehensive Safety Benchmark for Large Language Models [16]
4. UnsafeBench: Benchmarking Image Safety Classifiers on Real-World and AI-Generated Images [21]
5. Length-Controlled AlpacaEval: A Simple Way to Debias Automatic Evaluators [8]
6. GeoMLAMA: Geo-Diverse Commonsense Probing on Multilingual Pre-Trained Language Models [31]
7. Retrieval-augmented generation for knowledge-intensive nlp tasks [15]
8. SALMON: Self-Alignment with Instructable Reward Models [25]

Project Idea 3: Reducing Hallucinations of Large Vision Language Models

Pretrained large vision language models (LVLMs) have shown remarkable abilities, capable of fulfilling complex tasks and solving various vision-language problems according to human instructions. However, LVLMs still suffer from hallucinations, where models generate plausible yet incorrect outputs that are inconsistent with the visual information. This limits their reliability and restricts their application in sensitive scenarios such as medical settings. This project aims to reduce the hallucination issue of existing LVLMs. Your goal is use existing methods or propose creative new methods to reduce hallucinations, which will be measured on evaluation datasets. Additionally, you are encouraged to study how your proposed methods affect the overall behavior of LVLMs.

Tasks:

1. Select at least one LVLM for this project.
   (a) Options include proprietary models like GPT-4o or open-source models like LLaVA and InternVL.
2. Select at least one testset to evaluate hallucination.
   (a) You can use existing datasets such as POPE [17], AMBER [29], MMVP [27], HallusionBench [10], etc.
   (b) Alternatively, you can **create your own dataset** with at least 50 data entries (inputs + ground truth answers). To do that, first identify a unique scenario where hallucinations are either too common or cannot be tolerated, such as in medical applications. Such scenarios may not be sufficiently covered by existing general-domain datasets. Next, collect and annotate data points targeting the scenario, and try to ensure data diversity.
3. Design methods to reduce hallucination. Make sure to survey existing methods first. You can adopt and improve upon existing methods, or propose a creative new method. Some potential directions you can consider:
   (a) Emsemble features or predictions from multiple LVLMs;
   (b) Verify the consistency between model generation and visual information by prompting LVLMS, or by invoking external tools such as object detection models;
   (c) Design a decoding algorithm to enforce the consistency between model outputs and visual information;
   (d) Feel free to propose any other innovative ideas!
4. Evaluate the baseline LVLM and your proposed method on the evaluation datasets. Report the outcomes clearly. Ideally, your method can reduce hallucination on evaluation datasets. If not, try to explain why.
5. (Optional) Perform human evaluation on model outputs and compare human judgment with metric outcomes.
6. (Optional) In addition to evaluation datasets targeting hallucination, evaluate how your proposed method can be applied to general vision-language tasks. Provide qualitative examples to answer the following questions:
   (a) Is your proposed method applicable across different tasks? For example, can you apply your methods to broader tasks such as general VQA and instruction following?
   (b) In these broader tasks, does your method still help reduce hallucination?
   (c) If so, does this come at the cost of reduced instruction following or reasoning capabilities?
7. Prepare the project final presentation.
8. Draft a report with both quantitative and qualitative results.

Suggested Readings:

1. Hallucination of Multimodal Large Language Models: A Survey [3] This is a long and comprehensive survey, you can read it on an on-demand basis. Their github repo might be clearer and more organized: `https://github.com/showlab/Awesome-MLLM-Hallucination`

2. Evaluation benchmarks

   (a) Evaluating Object Hallucination in Large Vision-Language Models [17] (POPE)

   (b) AMBER: An LLM-free Multi-dimensional Benchmark for MLLMs Hallucination Evaluation [29]

   (c) HallusionBench: an advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models [10]

3. Methods in mitigating hallucination

   (a) Mitigating Object Hallucinations in Large Vision-Language Models through Visual Contrastive Decoding [14]

   (b) Mitigating Object Hallucination in Large Vision-Language Models via Classifier-Free Guidance [32]

   (c) Analyzing and mitigating object hallucination in large vision-language models [33]

Project Idea 4: Investigating Attention Mechanisms in LLMs for Downstream Applications

Attention mechanisms play a critical role in the success of Large Language Models (LLMs) by allowing them to focus on different parts of the input sequence dynamically. Understanding how attention operates within these models can provide valuable insights, improve model interpretability, and guide the development of more effective LLMs. This project aims to evaluate and compare the attention patterns in various LLMs across different tasks using various attention interpretation libraries and methodologies. The objective is to systematically analyze how attention functions across different models and interpretability methods, understand attention patterns in generative tasks, and observe how these patterns influence the models' performance on downstream applications.
Note: You are highly encouraged to explore your own ideas and design additional experiments beyond the ones outlined below.

Tasks:

1. Choose the libraries/tools for visualizing and analyzing attention mechanisms:
   (a) Example libraries: Inseq, Captum, LM Transparency Tool, etc.
   (b) Explore the features provided by these libraries and select the tools that suit your goals.
   (c) You must write code to use these libraries directly—do not rely on online demos or pre-existing interfaces.
2. Select **1-2 LLMs** that you will analyze:
   (a) Examples: LLaMA, BERT, etc.
   (b) Take your computational resources into account when choosing models.
3. Choose one or more interpretability methods to analyze attention:
   (a) Examples of interpretability methods: Gradient-based methods (e.g., Saliency Maps, Integrated Gradients), Attention Rollout, etc.
   (b) Provide reasoning for the selection of the method(s) based on the specific aspects you want to investigate (e.g., layer-wise attention, word-level importance, information flow, etc.).
4. Design or select a set of test datasets that will be used to probe the models' attention mechanisms. Ensure your dataset contains **at least 50 data entries**:
   (a) Example datasets: Text classification data (e.g., IMDb reviews for sentiment analysis), Extractive Summarization (e.g., CNN/Daily Mail), Question Answering (e.g., SQuAD), or Text Generation (e.g., prompts for creative writing).
   (b) Ensure the selected dataset is appropriate for the downstream application of interest.
5. Possible Experiments:
   - Experiment 1: Compare Attention Patterns Across Different LLMs
     (a) Analyze and compare how different LLMs allocate their attention to input tokens in similar contexts or tasks.
     (b) Identify patterns such as how much attention is paid to specific types of tokens (e.g., nouns, verbs, punctuation).
   - Experiment 2: Compare Attention Using Different Interpretability Methods
     (a) Evaluate the same LLM using multiple interpretability methods and compare how attention is captured by each method.
     (b) Identify similarities and differences in attention allocation across different interpretability approaches.

- Experiment 3: Analyze Attention Patterns During Text Generation
  (a) Investigate whether attention focuses more on specific tokens during generation (e.g., is the previous word always the most important?).
  (b) Analyze how attention shifts as the model generates longer text sequences.
- Experiment 4: Identify Attention Patterns in Downstream Tasks
  (a) Investigate how attention is distributed in downstream applications such as extractive summarization, sentiment analysis, or machine translation.
  (b) Determine which parts of the input receive the most attention and how this correlates with task performance.

6. Evaluate and Report Attention Patterns:
   (a) Create visualizations (e.g., heatmaps) to illustrate attention distributions.
   (b) Analyze and interpret the results, highlighting any notable trends or insights.
   (c) Example visualizations: Attention heatmaps, attention flow diagrams, or comparison charts between different methods/models.
7. Prepare the Final Presentation and Draft a Report:
   (a) Include quantitative (e.g., attention distribution statistics) and qualitative results (e.g., case studies, visualizations).
   (b) Discuss how attention patterns can inform the understanding of LLM performance on the chosen downstream applications.

Suggested Readings:

- Tools:
- Inseq: An Interpretability Toolkit for Sequence Generation Models [23]
- LM Transparency Tool: Interactive Tool for Analyzing Transformer Language Models [28]
- DecompX: Explaining Transformers Decisions by Propagating Token Decomposition [20]
- Captum: A unified and generic model interpretability library for PyTorch [13]
- Papers:
- Information Flow Routes: Automatically Interpreting Language Models at Scale [9]
- Towards Faithful Model Explanation in NLP: A Survey [19]
- Quantifying Attention Flow in Transformers [1]
- What Does BERT Look at? An Analysis of BERT's Attention [6]
- Attention is Not Only a Weight: Analyzing Transformers with Vector Norms [12]

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